Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Prof. Dr.-Ing. Bahman Azarhoushang	Email	bahman.azarhoushang@stw.de
Institute Name	KSF – Institute for Advanced Manufacturing / Steinbeis Transfer Centre NovaGrind			
Address	Hubbauerweg 26, 78048 Villingen-Schwenningen, Germany			

2. Technology Overview

Title of Proposed Technology	Edge-based Anomaly Detection and Workpiece Quality Prediction for CNC Turning Using Industrial AI Devices			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical·Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery, Robotics/ Automation Machinery	Cutting Machine Tools, Tools, Numerical Control (NC) Units, Measurement/Sensor Technologies for Automation		
Concept of Proposed Technology	This project aims to develop an industrial, edge-based AI system for real-time anomaly detection and workpiece quality prediction in CNC turning processes. An industrial edge device installed close to the machine tool acquires high-frequency sensor and control data (e.g., spindle current, acoustic emission, vibration, and OPC-UA signals). AI models deployed directly on the edge device analyze process behavior in real time to detect anomalies such as tool wear, tool breakage, chatter, and unstable cutting conditions [1,2]. In parallel, the system predicts resulting workpiece quality indicators, including surface roughness and dimensional deviations [3,4]. The system is designed for in-process operation with minimal latency and does not require continuous cloud connectivity, ensuring data security and robustness. Results are visualized for operators and production engineers, enabling early intervention, reduced scrap rates, and predictive process optimization.			

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Rationale for Joint R&D Project	CNC turning is a core precision manufacturing process widely used by Korean SMEs in the automotive, machinery, and high-value component industries. However, many SMEs still rely on reactive quality control and operator experience to detect process instabilities such as tool wear, chatter, or tool breakage. This leads to increased scrap rates, unplanned downtime, and limited process transparency. There is a strong industrial demand for in-process, real-time monitoring and quality prediction solutions that can be deployed economically on existing CNC machines [1,2]. The proposed technology requires the combination of advanced machining process knowledge, industrial AI algorithms, and robust edge-device engineering, which cannot be efficiently achieved by a single organization. The research institute contributes deep expertise in turning process mechanics, defect mechanisms, and data-driven modeling of machining quality, enabling the definition of meaningful anomaly patterns and quality indicators. In contrast, Korean SMEs and startups possess strong capabilities in industrial hardware development, embedded systems, CNC integration, and commercialization of manufacturing solutions tailored to shop-floor requirements. Joint R&D is essential to ensure that AI models are not developed in isolation, but are embedded into industrial-grade edge devices capable of high-frequency data acquisition, real-time processing, and reliable operation in harsh factory environments. The Korean partner's role in developing edge hardware, sensor integration, and interfaces to CNC controllers (e.g. Numerical Control units via OPC-UA) is critical for transforming research results into deployable products suitable for SME production lines. Furthermore, international collaboration enables validation of the system across different machine tool platforms and production conditions, increasing robustness and market applicability. By combining the research institute's process intelligence with the Korean partner's industrial platform and automation expertise, the project ensures rapid transition from laboratory-level AI models to commercially viable, edge-based monitoring solutions. This collaboration directly supports Korea's strategic goals in AI-driven advanced manufacturing and strengthens the competitiveness of Korean SMEs through reduced defects, improved productivity, and data-based decision making.
Key Words	1. Edge AI 2. Anomaly Detection 3. Workpiece Quality Prediction 4. CNC Turning 5. Smart Manufacturing

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	The objective of this R&D project is to develop an industrial, edge-based AI system for real-time monitoring, anomaly detection, and workpiece quality prediction in CNC turning processes. The system aims to enable early detection of process instabilities such as tool wear, tool breakage, chatter, and unstable cutting conditions, and to predict resulting workpiece quality during machining. Specific objectives include: • Development of an industrial edge device capable of high-frequency acquisition and real-time processing of multi-sensor and CNC control data • Design and implementation of AI models for reliable anomaly detection under varying cutting and production conditions • In-process prediction of workpiece quality indicators such as surface roughness and dimensional deviations • Integration with CNC controllers and automation systems using standard industrial interfaces (e.g. OPC-UA) • Visualization of monitoring results and predictive information to support operator decision-making • Validation of the system under real industrial CNC turning conditions, with a focus on SME applicability and retrofit capability			
R&D Target Specifications	• **Monitoring capability:** Real-time acquisition and processing of high-frequency sensor and CNC data (≥ 5 kHz) including spindle current, acoustic emission, vibration, and NC signals • **Anomaly detection performance:** Detection accuracy ≥ 90% for major turning-related anomalies (tool wear, tool breakage, chatter, unstable cutting) • **Quality prediction performance:** Prediction of surface roughness and dimensional deviation with an error ≤ ±10% compared to post-process measurements • **Latency:** End-to-end detection and prediction latency ≤ 100 ms to enable in-process intervention • **System architecture:** Fully edge-based operation without mandatory cloud connectivity, ensuring data security and robustness in factory environments • **CNC integration:** Compatibility with industrial CNC controllers via OPC-UA and standard automation interfaces • **Industrial readiness:** Retrofit installation on existing CNC turning machines and validation in an SME production environment (target TRL 7)			
R&D Plan	The proposed R&D activities are structured based on a **three-year** project framework, which provides sufficient time for systematic development, industrial integration, and validation of the proposed edge-based AI solution. The detailed timeline, task sequencing, and milestone allocation will be jointly defined and finalized in close coordination with the Korean industry partner prior to the project start, taking into account the partner's technical focus, available infrastructure, and commercialization strategy. All work packages, task contents, assignment of work-package leads and responsibilities, and technical specifications described are to be understood as a conceptual framework and will be closely coordinated with the Korean development partner, adapted as necessary, and further developed			

jointly to ensure optimal technical implementation and high industrial relevance.
The final R&D plan will be discussed, adjusted, and mutually agreed upon with the Korean SME or startup partner to ensure optimal efficiency, clear ownership, and strong industrial relevance throughout the project duration.

The project follows a **two-stream AI development strategy** implemented on an industrial edge platform.
Unsupervised learning is applied for **anomaly detection**, enabling the identification of unknown and previously unseen process abnormalities without labeled data.
Supervised learning is applied separately for **workpiece quality prediction**, where measured quality data are used to train predictive models.
Both streams share a common data acquisition and edge-computing infrastructure but are developed, validated, and deployed as **independent AI modules** to ensure robustness, explainability, and industrial usability. The system is designed for real-time, in-process operation with secure edge-based computation.

Work Packages (WP)
WP1 – Industrial Data Acquisition and Edge Platform Development
Lead (Optional-Suggestion): Korean SME / Startup
Objectives:
Establish a robust industrial edge platform for high-frequency data acquisition and real-time processing.
Key Tasks (SME):
- Design and development of industrial edge hardware for CNC turning machines
- Integration of multi-sensor inputs (spindle current, acoustic emission, vibration, temperature)
- Implementation of real-time data acquisition (≥ 5 kHz)
- CNC controller integration via OPC-UA and NC interfaces
- Data buffering, synchronization, and preprocessing on the edge device

Methodology:
- Embedded system design
- Real-time operating system (RTOS) or low-latency Linux environments
- Industrial communication protocols

Milestone M1:
Operational edge device acquiring synchronized multi-sensor and CNC data in real time.

WP2 – Process Understanding and Feature Engineering
Lead (Optional-Suggestion): Research Institute
Objectives:
Translate turning process knowledge into meaningful data features for AI models.
Key Tasks (Research Institute):
- Identification of process-relevant signals and frequency bands
- Feature extraction for cutting stability, tool condition, and process dynamics
- Definition of reference “normal operation” states
- Feature standardization and normalization strategies

Methodology:
- Machining process analysis
- Signal processing (time–frequency analysis, statistical descriptors)
- Physics-informed feature engineering

Milestone M2:
Validated feature sets suitable for anomaly detection and quality prediction models.

WP3 – Anomaly Detection Using Unsupervised Learning
Lead (Optional-Suggestion): Research Institute
Objectives:
Develop AI models capable of detecting unknown and unexpected process anomalies.
Key Tasks (Research Institute):
- Selection of unsupervised learning approaches (e.g. autoencoders, clustering, isolation methods)
- Training using data from stable turning processes
- Detection of deviations from learned normal behavior
- Definition of anomaly severity levels
- Explainability of detected anomalies for operators

Methodology:
- Unsupervised machine learning
- Model validation using induced process disturbances
- Statistical performance evaluation (false positives, sensitivity)

Milestone M3:
Unsupervised anomaly detection module capable of identifying tool wear, chatter, and abnormal cutting behavior.

WP4 – Workpiece Quality Prediction Using Supervised Learning
Lead (Optional-Suggestion): Research Institute
Objectives:
Predict final workpiece quality based on in-process data.
Key Tasks (Research Institute):
- Definition of target quality metrics (surface roughness, dimensional deviation)
- Acquisition and labeling of quality measurement data
- Training of supervised learning models
- Correlation of in-process signals with post-process quality results
- Model generalization across different cutting conditions

Methodology:
- Supervised machine learning
- Regression and ensemble methods
- Cross-validation and uncertainty estimation

Milestone M4:
Validated quality prediction models achieving target accuracy under industrial conditions.

WP5 – Edge Deployment and System Integration
Lead (Optional-Suggestion): Korean SME / Startup
Objectives:
Deploy AI models on the industrial edge device and integrate into CNC environments.
Key Tasks (SME):
- Optimization of AI models for edge deployment
- Real-time inference implementation
- Integration of anomaly and quality outputs into operator interfaces
- Alarm management and visualization
- System stability, cybersecurity, and data integrity measures

Methodology:

- Edge AI optimization
- Software–hardware co-design
- Industrial HMI development

Milestone M5:
Fully integrated edge-based monitoring system operating in real time on a CNC turning machine.

WP6 – Industrial Validation and Pilot Operation
Lead (Optional-Suggestion): Joint
Objectives:
Validate system performance and industrial applicability.
Key Tasks (Joint):

- Pilot deployment on SME production machines
- Evaluation of anomaly detection accuracy and prediction reliability
- Assessment of scrap reduction and downtime avoidance
- Operator feedback and system refinement
- TRL assessment and commercialization readiness evaluation

Methodology:

- Industrial field testing
- Comparative studies (with/without AI support)
- Iterative improvement cycles

Milestone M6:
TRL 7 system validated in a production environment.

Work Packages Overview 3-Year Project Framework)
The project is structured within a **three-year R&D framework**. The detailed scheduling, duration, and sequencing of individual work packages will be **jointly defined and finalized with the Korean industry partner** prior to the beginning of the project.

- **WP1:** Development of the industrial edge platform and data acquisition infrastructure
- **WP2:** Feature engineering and process-relevant data interpretation
- **WP3:** Development of unsupervised learning methods for anomaly detection
- **WP4:** Development of supervised learning models for workpiece quality prediction
- **WP5:** System integration, edge deployment, and CNC connectivity
- **WP6:** Industrial validation, pilot operation, and system refinement

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the **scientific and methodological foundations** of the project, focusing on machining process intelligence and AI model development. Its main roles and responsibilities include: • Definition and modeling of CNC turning process behavior, including identification of relevant process variables, defect mechanisms, and quality indicators • Development of signal processing and feature engineering methods tailored to turning processes • Design and implementation of **unsupervised learning models** for real-time anomaly detection • Design and implementation of **supervised learning models** for workpiece quality prediction based on labeled measurement data • Validation and performance evaluation of AI models using experimental and industrial data • Development of explainable AI concepts to support operator understanding and trust • Scientific coordination of AI-related work packages and contribution to system architecture design • Documentation of methodologies, validation results, and technical guidelines • Support of pilot operation, result interpretation, and refinement of algorithms based on industrial feedback The research institute will ensure that all AI methods are **technically sound, explainable, and transferable** to industrial environments, while maintaining close coordination with the Korean SME partner.
Expected R&R of Korean SME Partner	The Korean SME or startup partner will be responsible for the **industrial realization, system integration, and commercialization-oriented development** of the proposed technology. Its main roles and responsibilities include: • Design and development of **industrial edge hardware** suitable for installation on CNC turning machines • Integration of multi-sensor systems and high-frequency data acquisition on the edge device • Implementation of real-time data processing, buffering, and system stability mechanisms • Integration with CNC controllers and automation systems using industrial interfaces (e.g. OPC-UA, NC units) • Deployment of AI models provided by the research institute onto the edge device, including optimization for real-time inference • Development of operator-oriented user interfaces, dashboards, and alarm systems • Implementation of cybersecurity, data protection, and access control functions suitable for factory environments • Execution of pilot operation and industrial validation in SME production settings • Collection of operational feedback and support of system refinement • Preparation of commercialization strategies, product scaling, and customization for different customer requirements The Korean SME partner will ensure that the developed solution meets **industrial robustness, usability, and market readiness requirements**, enabling effective deployment in SME manufacturing environments.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcome of this joint R&D project is a **validated, edge-based AI system** for real-time anomaly detection and workpiece quality prediction in CNC turning processes, demonstrated under industrial conditions. The project will deliver a deployable prototype capable of monitoring machining processes in real time, detecting process instabilities at an early stage, and predicting final workpiece quality during machining. From a technological perspective, the project will establish a **robust industrial AI framework** that combines high-frequency sensor data, CNC control signals, and advanced machine-learning models deployed directly on edge devices. The clear separation between unsupervised anomaly detection and supervised quality prediction ensures high reliability, adaptability to varying production conditions, and reduced dependency on large labeled datasets. The developed methods will be transferable to other precision machining processes and machine tool platforms. From an industrial and economic perspective, the solution is expected to significantly **reduce scrap rates, unplanned downtime, and quality-related rework** in CNC turning operations. By enabling early intervention and predictive process optimization, SMEs can improve productivity, stabilize production quality, and reduce dependence on highly experienced operators. The edge-based architecture ensures data security and allows retrofit deployment on existing CNC machines, which is particularly important for SMEs with heterogeneous equipment. The project will also strengthen the **commercialization potential** of Korean SMEs and startups by delivering an AI-enabled industrial monitoring product with clear market differentiation. The developed edge device and software platform can be scaled and customized for different customer requirements, supporting sustainable business growth and international competitiveness. At a broader level, the project contributes to Korea's strategic objectives in **AI-driven advanced manufacturing and smart factory technologies**, supporting digital transformation of the manufacturing sector. The collaboration enhances international R&D cooperation, accelerates technology transfer from research to industry, and reinforces Korea's position as a leader in intelligent, high-value manufacturing systems.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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